

14. COMPUTER SCIENCE (Code 083)

Learning Objectives:

1. To develop logic for problem solving
2. To understand the concept of Object Oriented Methodology
3. To implement Object Oriented Programming using C++
4. To understand the concept of working with Relational Database
5. To understand the basic concept of algebra of logic
6. To understand and explore the world of communication and networks
7. To understand the concept of Web Services
8. To understand localisation issues

Competencies:

The student will be proficient in the following:

1. Identification of a Computer System
2. Problem Solving using object oriented programming
3. Designing an efficient logic using object oriented approach for solution development handling
4. Database handling
5. Logic Circuit designing
6. Network concepts and Web Services

Class XI (Theory)

Duration: 3 hours

Total Marks: 70

Unit No.	Unit Name	Marks
1.	COMPUTER FUNDAMENTALS	10
2.	PROGRAMMING METHODOLOGY	10
3.	INTRODUCTION TO C++	15
4.	PROGRAMMING IN C++	35
		70

UNIT 1: COMPUTER FUNDAMENTALS

Evolution of computers; Basics of computer and its operation: Functional Components and their inter-connections, concept of Booting.

Software Concepts:

Types of Software - System Software, Utility Software and Application Software;

System Software: Operating System, Compilers, Interpreters and Assembler;

Utility Software : Anti Virus, File Management tools, Compression tools and Disk Management tools (Disk Cleanup, Disk Defragmenter, Backup);

Application Software as a tool: Word Processor, Presentation tools, Spreadsheet Package, Database Management System; Business software (for example: School Management System, Inventory Management System, Payroll System, Financial Accounting, Hotel Management, and Reservation System);

Operating System : Need for operating system, Functions of Operating System (Processor Management, Memory Management, File Management and Device Management), Types of operating system – Interactive (GUI based), Time Sharing, Real Time and Distributed; Commonly used operating systems: LINUX, Windows, BhartiOO, Solaris, UNIX;

Illustration and practice of the following tasks using any one of the above Operating Systems:

- Opening / Closing Windows
- Creating / Moving / Deleting Files / Folders
- Renaming Files / Folders
- Switching between Tasks

Number System : Binary, Octal, Decimal, Hexadecimal and conversion between two different number systems;

Internal Storage encoding of Characters: ASCII, ISCII (Indian scripts Standard Code for Information Interchange), and UNICODE;

Microprocessor : Basic concepts, Clock speed (MHz, GHz), 16 bit, 32 bit, 64 bit processors; Types – CISC, RISC;

Memory Concepts :

Units : Byte, Kilo Byte, Mega Byte, Giga Byte, Tera Byte, Peta Byte

Primary Memory : Cache, RAM, ROM,

Secondary Memory : Hard Disk Drive, CD / DVD Drive, Pen Drive, Blue Ray Disk;

Input Output Ports / Connections: Serial, Parallel and Universal Serial Bus, PS-2 Port, Infrared port, Bluetooth.

UNIT 2: PROGRAMMING METHODOLOGY

General Concepts; Modular approach; Clarity and Simplicity of Expressions, Use of proper Names for identifiers, Comments, Indentation; Documentation and Program Maintenance; Running and Debugging programs, Syntax Errors, Run-Time Errors, Logical Errors;

Problem Solving Methodology and Techniques: Understanding of the problem, Identifying minimum number of inputs required for output, Step by step solution for the problem, breaking down solution into simple steps, Identification of arithmetic and logical operations required for solution, Using Control Structure: Conditional control and looping (finite and infinite);

UNIT 3: INTRODUCTION TO C++

Getting Started:

C++ character set, C++ Tokens (Identifiers, Keywords, Constants, Operators), Structure of a C++ Program (include files, main function); Header files – iostream.h, iomanip.h; **cout, cin**; Use of I/O operators (<< and >>), Use of endl and setw(), Cascading of I/O operators, Error Messages; Use of editor, basic commands of editor, compilation, linking and execution; standard input/output operations from C language: gets(), puts() of stdio.h header file;

Data Types, Variables and Constants:

Concept of Data types; Built-in Data types: **char, int, float** and **double**; Constants: Integer Constants, Character Constants (Backslash character constants - \n, \t), Floating Point Constants, String Constants; Access modifier: **const**; Variables of built-in data types, Declaration/Initialisation of variables, Assignment statement; Type modifier: signed, unsigned, long;

Operators and Expressions:

Operators: Arithmetic operators (-, +, *, /, %), Unary operator (-), Increment and Decrement Operators (—, ++), Relational operators (>, >=, <, <=, ==, !=), Logical operators (!, &&, ||), Conditional operator: <condition>?<if true>:<else>; Precedence of Operators; Expressions; Automatic type conversion in expressions, Type casting; C++ shorthand's (+=, -=, *=, /=, %=);

UNIT 4: PROGRAMMING IN C++

Flow of control:

Conditional statements: **if-else**, Nested **if**, **switch..case..default**, Nested **switch..case**, break statement (to be used in switch..case only); Loops: **while**, **do - while** , **for** and Nested loops;

String Functions:

Header File: string.h

Function: **isalnum(), isalpha(), isdigit(), islower(), isupper(), tolower(), toupper()**;

Character Functions:

Header File: ctype.h

Functions: **isalnum(), isalpha(), isdigit(), islower(), isupper(), tolower(), toupper(), strcpy(), strcat(), strlen(), strcmp(), strcmpi()**;

Mathematical Functions:

Header File-math.h, stdlib.h;

Functions: **fabs(), log(), log10(), pow(), sqrt(), sin(), cos(), abs()**,

Other Functions:

Header File- `stdlib.h`;

Functions: **`randomize()`**, **`random()`**;

User Defined Functions:

Defining a function; function prototype, Invoking/calling a function, passing arguments to function, specifying argument data types, default argument, constant argument, call by value, call by reference, returning values from a function, calling functions with arrays, scope rules of functions and variables; local and global variables;

Structured Data Type: Array

Declaration/initialisation of One-dimensional array, Inputting array elements, Accessing array elements, Manipulation of Array elements (sum of elements, product of elements, average of elements, linear search, finding maximum/minimum value);

Declaration/Initialization of a String, string manipulations (counting vowels/consonants/digits/special characters, case conversion, reversing a string, reversing each word of a string);

Two-dimensional Array :

Declaration/initialisation of a two-dimensional array, inputting array elements Accessing array elements, Manipulation of Array elements (sum of row element, column elements, diagonal elements, finding maximum/minimum values);

User-defined Data Types

Need for User defined data type:

Defining a symbol name using `typedef` keyword and defining a macro using `#define` directive;

Structures:

Defining a Structure, Declaring structure variables, Accessing structure elements, Passing structure of Functions as value and reference argument/parameter, Function returning structure, Array of structures, passing an array of structure as an argument/ a parameter to a function.

Class XI (Practical)

Duration: 3 hours

Total Marks: 30

1. Programming in C++

10

One programming problem in C++ to be developed and tested in Computer during the examination. Marks are allotted on the basis of following:

Logic : 5 Marks

Documentation/Indentation : 2 Marks

Output presentation : 3 Marks

2. Project Work

10

Problems related to String, Number and Array manipulation;

General Guidelines : Initial Requirement, developing an interface for user (it is advised to use text based interface screen), developing logic for playing the game and developing logic for scoring points

1. Memory Game : A number guessing game with application of 2 dimensional arrays containing randomly generated numbers in pairs hidden inside boxes.
2. Cross 'N Knots Game : A regular tic-tac-toe game
3. Hollywood/Hangman: A word Guessing game
4. Cows 'N Bulls : A word/number Guessing game

Similar projects may be undertaken in other domains

(As mentioned in general guidelines for projects, given at the end of the curriculum in a group of 1-2 students)

3. Practical File

05

Must have minimum 15 programs from the topics covered in class XI course.

- 5 Programs on Control structures
- 4 Programs on Array Manipulations
- 4 Programs on String Manipulations
- 2 Programs on structure manipulations

4. Viva Voce

05

Viva will be asked from syllabus covered in class XI and the project developed by student.

Class XII (Theory)

Duration: 3 hours

Total Marks: 70

Unit No.	Unit Name	Marks
1.	PROGRAMMING IN C++	30
2.	DATA STRUCTURE	14
3.	DATABASE AND SQL	8
4.	BOOLEAN ALGEBRA	8
5.	COMMUNICATION AND OPEN SOURCE CONCEPTS	10
		70

UNIT 1: PROGRAMMING IN C++

REVIEW: C++ covered In Class -XI,

Object Oriented Programming:

Concept of Object Oriented Programming – Data hiding, Data encapsulation, Class and Object, Abstract class and Concrete class, Polymorphism (Implementation of polymorphism using Function overloading as an example in C++); Inheritance, Advantages of Object Oriented Programming over earlier programming methodologies,

Implementation of Object Oriented Programming concepts in C++:

Definition of a class, Members of a class - Data Members and Member Functions (methods), Using Private and Public visibility modes, default visibility mode (private); Member function definition: inside class definition and outside class definition using scope resolution operator (::); Declaration of objects as instances of a class; accessing members from object(s), Array of type class, Objects as function arguments - pass by value and pass by reference;

Constructor and Destructor:

Constructor: Special Characteristics, Declaration and Definition of a constructor, Default Constructor, Overloaded Constructors, Copy Constructor, Constructor with default arguments;

Destructor: Special Characteristics, Declaration and definition of destructor;

Inheritance (Extending Classes):

Concept of Inheritance, Base Class, Derived Class, Defining derived classes, protected visibility mode; Single level inheritance, Multilevel inheritance and Multiple inheritance, Privately derived, Publicly derived and Protectedly derived class, accessibility of members from objects and within derived class(es);

Data File Handling:

Need for a data file, Types of data files – Text file and Binary file;

Text File: Basic file operations on text file: Creating/Writing text into file, Reading and manipulation of text from an already existing text File (accessing sequentially);

Binary File: Creation of file, Writing data into file, Searching for required data from file, Appending data to a file, Insertion of data in sorted file, Deletion of data from file, Modification of data in a file;

Implementation of above mentioned data file handling in C++;

Components of C++ to be used with file handling:

Header file: fstream.h; ifstream, ofstream, fstream classes;

Opening a text file in **in**, **out**, and **app** modes;

Using cascading operators for writing text to the file and reading text from the file; **open()**, **get()**, **put()**, **getline()** and **close()** functions; Detecting end-of-file (with or without using **eof()** function);

Opening a binary file using **in**, **out**, and **app** modes;

open(), **read()**, **write()** and **close()** functions; Detecting end-of-file (with or without using **eof()** function); **tellg()**, **tellp()**, **seekg()**, **seekp()** functions

Pointers:

Declaration and Initialization of Pointers; Dynamic memory allocation/deallocation operators: **new**, **delete**; Pointers and Arrays: Array of Pointers, Pointer to an array (1 dimensional array), Function returning a pointer, Reference variables and use of alias; Function call by reference. Pointer to structures: Deference operator: *, ->; self referencial structures;

UNIT 2: DATA STRUCTURES

Arrays:

One and two Dimensional arrays: Sequential allocation and address calculation;

One dimensional array: Traversal, Searching (Linear, Binary Search), Insertion of an element in an array, deletion of an element from an array, Sorting (Insertion, Selection, Bubble sort), concatenation of two linear arrays, merging of two sorted arrays;

Two-dimensional arrays: Traversal, Finding sum/difference of two NxM arrays containing numeric values, Interchanging Row and Column elements in a two dimensional array;

Stack (Array and Linked implementation of Stack):

Operations on Stack (PUSH and POP) and its Implementation in C++, Converting expressions from INFIX to POSTFIX notation and evaluation of Postfix expression;

Queue: (Circular Array and Linked Implementation):

Operations on Queue (Insert and Delete) and its Implementation in C++.

UNIT 3: DATABASES AND SQL

Database Concepts:

Relational data model: Concept of domain, tuple, relation, key, primary key, alternate key, candidate key;

Relational algebra: Selection, Projection, Union and Cartesian product;

Structured Query Language:

General Concepts: Advantages of using SQL, Data Definition Language and Data Manipulation Language;

Data types: NUMBER, CHARACTER, DATE;

SQL commands:

CREATE TABLE, DROP TABLE, ALTER TABLE, UPDATE...SET..., INSERT, DELETE;

SELECT, DISTINCT, FROM, WHERE, IN, BETWEEN, GROUP BY, HAVING, ORDER BY;

SQL functions: SUM, AVG, COUNT, MAX and MIN;

Note: Implementation of the above mentioned commands could be done on any SQL supported software on one or two tables.

UNIT 4: BOOLEAN ALGEBRA

Binary-valued Quantities, Boolean Variable, Boolean Constant and Boolean Operators: AND, OR, NOT; Truth Tables; Closure Property, Commutative Law, Associative Law, Identity law, Inverse law, Principle of Duality, Idempotent Law, Distributive Law, Absorption Law, Involution law, DeMorgan's Law and their applications;

Obtaining Sum of Product (SOP) and Product of Sum (POS) form from the Truth Table, Reducing Boolean Expression (SOP and POS) to its minimal form, Use of Karnaugh Map for minimisation of Boolean expressions (up to 4 variables);

Basic Logic Gates (NOT, AND, OR, NAND, NOR) and their use in circuits.

UNIT 5: COMMUNICATION AND OPEN SOURCE CONCEPTS

Evolution of Networking: ARPANET, Internet, Interspace;

Different ways of sending data across the network with reference to switching techniques;

Data Communication terminologies:

Concept of Channel, Baud, Bandwidth (Hz, KHz, MHz) and Data transfer rate (bps, kbps, Mbps, Gbps, Tbps);

Transmission media:

Twisted pair cable, coaxial cable, optical fiber, infrared, radio link, microwave link and satellite link.

Network devices:

Modem, RJ45 connector, Ethernet Card, Hub, Switch, Gateway;

Network Topologies and types:

Bus, Star, Tree; Concepts of LAN, WAN, MAN

Network Protocol:

TCP/IP, File Transfer Protocol (FTP), PPP, Level-Remote Login (Telnet), Internet, Wireless/Mobile Communication, GSM, CDMA, WLL, 3G, SMS, Voice mail, Application Electronic Mail, Chat, Video Conferencing;

Network Security Concepts:

Threats and prevention from Viruses, Worms, Trojan horse, Spams

Use of Cookies, Protection using Firewall;

India IT Act, Cyber Law, Cyber Crimes, IPR issues, Hacking.

Web Servers;

Hyper Text Markup Language (HTML), extensible Markup Language (XML); Hyper Text Transfer Protocol (HTTP); Domain Names; URL; Protocol Address; Website, Web browser, Web Servers; Web Hosting, Web Scripting – Client side (VB script, Java Script, PHP) and Server side (ASP, JSP, PHP)

Open Source Terminologies:

Open Source Software, Freeware, Shareware, Proprietary software, FLOSS, GNU, FSF, OSI;

Class XII (Practicals)

Duration: 3 hours

Total Marks: 30

1. Programming in C++

10

One programming problem in C++ to be developed and tested in Computer during the examination. Marks are allotted on the basis of following:

Logic	:	5 Marks
Documentation/Indentation	:	2 Marks
Output presentation	:	3 Marks

Notes: The types of problems to be given will be of application type from the following topics

- Arrays (One dimensional and two dimensional)
- Array of structure
- Stack using arrays and linked implementation
- Queue using arrays (circular) and linked implementation
- Binary File operations (Creation, Displaying, Searching and modification)
- Text File operations (Creation, Displaying and modification)

2. SQL Commands

05

Five Query questions based on a particular Table/Relation to be tested practically on Computer during the examination. The command along with the result must be written in the answer sheet.

3. Project Work

05

The project has to be developed in C++ language with Object Oriented Technology and also should have use of Data files. (The project is required to be developed in a group of 1-2 students)

- Presentation on the computer
- Project report (Listing, Sample, Outputs, Documentation)
- Viva

4. Practical File

05

Must have minimum 20 programs from the following topics

- Arrays (One dimensional and two dimensional, sorting, searching, merging, deletion & insertion of elements)
- Arrays of structures, Arrays of Objects
- Stacks using arrays and linked implementation
- Queues using arrays (linear and circular) and linked implementation
- File (Binary and Text) operations (Creation, Updation, Query)
- Any computational based problems

15 SQL commands along with the output based on any table/relation: 3 Marks

5. Viva Voce

05

Viva will be asked from syllabus covered in class XII and the project developed by student.

GUIDELINES FOR PROJECTS (Class XI and XII)

1. Preamble

- 1.1 The academic course in Computer Science includes one Project in each year. The Purpose behind this is to consolidate the concepts and practices imparted during the course and to serve as a record of competence.
- 1.2 A group of two students/three students as team may be allowed to work on one project.

2. Project content

- 2.1 Project for class XI can be selected from the topics mentioned in syllabus or domains on the similar lines.
- 2.2 Project for class XII should ensure the coverage of following areas of curriculum:
 - a. Problem Solving
 - b. Data Structure
 - c. Object Oriented Programming in C++
 - d. Data File Handling

Theme of the project can be

- Any subsystem of a System Software or Tool
 - Any Scientific or a fairly complex algorithmic situation.
 - Business oriented problems like Banking, Library information system, Hotel or Hospital management system, Transport query system
 - Quizzes/Games;
 - Tutor/Computer Aided Learning Systems
- 2.3 The aim of the project is to highlight the abilities of algorithmic formulation, modular programming, optimized code preparation, systematic documentation and other associated aspects of Software Development.
 - 2.4 The assessment would be through the project demonstration and the Project Report, which should portray Programming Style, Structured Design, Minimum Coupling, High Cohesion, Good documentation of the code to ensure readability and ease of maintenance.

Suggested Reference Books

Computer Organisation and Boolean Algebra

1. Rajaraman, FUNDAMENTALS OF COMPUTERS 4th Edition, Prentice Hall of India.
2. Peter Norton, INTRODUCTION TO COMPUTER 4th Edition, Tata McGraw Hill

3. J. Shelly & Roger Hunt, COMPUTER STUDIES, Wheeler's Publication.
4. C.S. French, COMPUTER STUDIES, Arnold Publishers.
5. Thomas C. Bartee, DIGITAL COMPUTER FUNDAMENTALS, McGraw Hill International.

Problem Solving and Programming in C++

Note: Prior knowledge of C is not required in the learning of C++, even though reference about C are made in some of the books.

1. Robert Lofore, OBJECT ORIENTED PROGRAMMING IN TURBO C++, Galgotia Publications Pvt. Ltd.
2. David Parsons, OBJECT ORIENTED PROGRAMMING WITH C++, BPB Publications.
3. Bjarne Stroustrup, THE C++ PROGRAMMING LANGUAGE, Addison Wesley.

Data Structures

1. M.A. Weiss, Data Structures and Algorithm Analysis in C++. the Benjamin/Cummings Pub. Co., Inc.
2. Scott Rober Ladd, C++ COMPONENTS AND ALGORITHMS, BPB Publications.
3. Sartaj & Sahni, Fundamentals of Data Structure, Galgotia Book Source

Database Management System and SQL

1. C.J. Data, DATABASE PRIMER, Addison Wesley.
2. Martin Gruber, UNDERSTANDING SQL, BPB Publications.
3. Sheldon M. Dunn x Base Cross Reference Handbook, First Authorised Asian Edition 93, Tech. Publications Pvt. Ltd.

Computer Network

1. A.S. Tanenbaum, Computer Network 4th Edition, Prentice Hall of India P.Ltd.
2. Williams Stalling, Data Communication and Networks 5th Edition, Prentice Hall of India P. Ltd.
3. Hancock, Network Concept and Architectures, BPB Publications.

15. INFORMATICS PRACTICES (Code 065)

Learning Objectives:

1. To understand the application development environment.
2. To gain programming Skills in GUI Programming Tool and Database Creation in RDBMS.
3. To design, program and develop database application using GUI Programming Tool and RDBMS.
4. To learn database connectivity using Visual Basic as Front-end tool.
5. To develop ability to use the Open Source Technology.

Competencies:

1. Student will become familiar with Application Development
2. Student will be able to develop & debug programs Independently.
3. Student can use SQL for storing and retrieving data from the RDBMS.
4. Ability to arrive at a normalized design of tables and other database objects in RDBMS.
5. Student will be able to develop a Client Server Application using Front end and Back end tools.

Class XI (Theory)

Duration: 3 hours

Total Marks: 70

Unit No.	Unit Name	Marks
1.	COMPUTER SYSTEM AND BUSINESS APPLICATIONS	10
2.	INTRODUCTION TO PROGRAMMING	30
3.	RELATIONAL DATABASE MANAGEMENT SYSTEM	30
		70

UNIT 1: COMPUTER SYSTEM AND BUSINESS APPLICATIONS

Evolution of computers; Basics of computer and its operation: Functional Components and their inter-connections, concept of Booting;

Hardware concepts:

Diagram illustrating main parts of computers;

Central Processing Unit (CPU): Arithmetic Logic Unit (ALU), Control Unit, Memory Unit (RAM - Random Access Memory & ROM - Read Only Memory)

Role of Input, Processing and Output Devices in a computer system

Input devices: Keyboard, Mouse, Light pen, Touch Screens, Graphics Tablets, Joystick, Mic, MICR, OCR, Scanner, Smart Card reader, Barcode reader, Biometric sensor, web camera, digital camera; Output Devices: Monitor/Visual Display Unit (VDU), Printer (Dot Matrix Printer, Desk jet/ Ink jet/ Bubble jet Printer, Laser Printer), Plotter, Speaker,

Secondary Storage Devices: Floppy Disk, Hard Disk, Compact Disk, Magnetic Tape, Digital Video Disk (DVD), Zip Drive; Units of Memory: Bit (Binary Digit), Byte, Kilobyte, Megabyte, Gigabyte.

Software Concepts:

Types of Software: System Software, Utility Software and Application Software.

System Software: Operating System, Language Compilers, Interpreters and Assembler;

Operating System: Need of operating systems, Functions of Operating System Types of operating system.

Utility Software: Compression tools, Anti Virus, File Management tools and Disk Management tools;
Application Software as a tool: Word Processor, Presentation Tool, Spreadsheet Package, Database Management System; Business software (for example: Inventory Management System, Payroll System, Financial Accounting, Hotel Management, and Reservation System);
Development of programming languages - Machine Language, Assembly Language, High Level Language (BASIC, COBOL, FORTRAN, PASCAL, C++); GUI based languages - Visual Basic, Visual C++; C#, Java, vb.net.

GUI Operating System

Important: *Students/Teachers can also perform similar operation on any operating system. It is advised that the teacher while using any one operating system, give a demonstration of equivalent features for the other operating system.*

Windows

General features, Elements of Desktop - Taskbar, Icon, Start button, Shortcuts, Folder, Recycle Bin, My Computer;

Start Menu: Program, Documents, Settings, Find/Search, Help, Run, Shut Down/Logoff;

Customization of Taskbar, start menu, Display properties (Wallpaper, Font Settings, Color Settings, Screen Savers);

Program Menu: Accessories - Calculator, Notepad, Paint, Word pad, Entertainments (CD Player, Sound Recorder, Media Player, Volume Controller); Internet Browsers – Mozilla Firefox, Internet Explorer, Netscape Navigator.

Control Panel: Add new hardware; Add new Software, Printer Installation, Date/Time, Mouse, and Regional Settings;

Documentation

Purpose of using word processing software, opening a new/existing document, closing a document, typing in a document, saving a document, print preview, printing a document, setting up of page as per the specifications, selecting a portion of document, copying selected text, cutting selected text, pasting selected text; changing font, size, style, color of text; Inserting symbol; Formatting: Alignment – Left, Right, Center; Justification;

Industries and Business Computing: Types of Industries (Production, Shipping, Travel, Hotel, Insurance, Construction, Automobile), Applications of Business Computing in Industries.

UNIT 2: INTRODUCTION TO PROGRAMMING

Programming Methodology:

General Concepts; Modular approach; Stylistic Guidelines: Clarity and Simplicity of Expressions, Names, Comments, Indentation; Documentation and Program Maintenance; Running and Debugging programs, Syntax Errors, Run-Time Errors, Logical Errors;

Problem Solving Methodology and Techniques: Understanding of the problem, Identifying minimum number of inputs required for output, Step by step solution for the problem, breaking down solution into simple steps, Identification of arithmetic and logical operations required for solution, Using Control Structure: Conditional control and looping (finite and infinite);

Programming Tool: Visual Basic

Introduction to Programming – Modular Programming, Object Oriented Programming, Event Driven Programming;

About Visual Basic (Object Based Programming Language), Rapid Application Development using Visual Basic; Concept of Project in Visual Basic, VB Project Options - Standard EXE, ActiveX DLL, ActiveX EXE, ActiveX Control, ActiveX Document DLL, ActiveX Document EXE, Addin, VB Application Wizard, IIS Application, DHTML Application;

Getting Familiar with Visual Basic User Interface - Pull-Down menus, Toolbar, Toolbox, Project Explorer, Properties Window, Form Layout window, Form, Immediate window;

Opening and Closing windows, Resizing and moving windows, Docking windows; Quitting Visual Basic;

Visual Basic Tool Box (Standard Window Controls) - Pointer, Picture Box, Label, Text Box, Frame, Command Button, Check Box, Option Button, Combo Box, List Box, Horizontal Scrollbar, Vertical Scrollbar, Timer, Drive List box, Directory List box, File List box, Shape, Line, Image, Data, OLE; Object Naming Conventions, Event Procedures;

Programming Fundamentals

Data Types: Integer, Long, Single, Double, Currency, String, Byte, Boolean, Date, Object, Variant; Variables: Need to use variable, Declaring Variables, Variable Naming Convention, Assigning value to Variables, Data Types of variable, Scope and lifetime of Variables (Public and Private);

Control Structures:

Decision Structure – IF, IF-THEN-ELSE, Select Case;

Looping Structure- Do While...Loop, Do...Loop While, For...Next, For Each...Next;

Menu Editor: Concept of Menus, Shortcut menus and Popup menus Designing Menu System, Menu Editor Dialog Box Options (Name, Index, Shortcut, HelpContextID, NegotiatePosition, Checked, Enabled, Visible, WindowList, Right Arrow, Left Arrow, Up Arrow, Down Arrow, Menu List, Next, Insert, Delete, OK, Cancel), To Create Menu Controls in the Menu Editor, Menu Naming Conventions, Setting the Name Property, Creating a Menu Control Array, Creating Sub Menus, Separating Menu Controls, Assigning Access Keys and Shortcut Keys, Controlling Menus at Runtime-Enabling and Disabling Menu Commands, Displaying a Checkmark on a Menu Control, Making a Menu Control Invisible, Adding Menu Control at Runtime, Displaying Pop-Up Menu;

General Controls (Advance): Image List, Common Dialog Box, ADO DC, DB Combo, Media Player Control, DB Grid;

Adding a Toolbar: Creating an Image List, Adding Images to the Toolbar, To Add Code for the Toolbar Buttons;

Adding Status Bar: Adding Status Bar panels, Adding Time on the panel.

Dialog Boxes: Pre-defined dialog box, Custom dialog box;

UNIT 3: RELATIONAL DATABASE MANAGEMENT SYSTEM

Database Management System

Introduction to database concepts: relation/Table, attribute, Tuple / Rows, field, Data, Concept of String, Number and Date values, Data type and Data Integrity (Domain and Referential

Integrity). Candidate key, Alternate key, Primary Key, Foreign Keys; Data Normalization-first, second, third, BCNF normal form;

Examples of Commercially available Database Management System's (Back-End) – Oracle, MS-SQL Server, DB2, MySQL, Sybase, INGRES.

Examples of Front End Software's: Oracle Developer, Visual Basic, Visual C++, Power Builder, Delphi;

RDBMS Tool :

Interface with Oracle, Login Screen, Entering Name and Password; Classification of SQL Statements: DML (SELECT, INSERT, UPDATE, DELETE), DDL (CREATE, DROP, ALTER, RENAME, TRUNCATE), DCL (GRANT, REVOKE), TCL (COMMIT, ROLLBACK); SQL SELECT Statement: SQL SELECT statement, Selecting All the Columns, Selecting Specific Column, Column Heading Default, Using Arithmetic Operators, Operator Precedence, Significance of NULL value, NULL values in Arithmetic Expressions, Defining and using Column Alias, Concatenation Operator (||), Duplicate rows and their Elimination (DISTINCT keyword), Role of SQL and SQL*Plus in interacting with RDBMS, Displaying Table Structure (DESC command);

SELECT Statement Continued: Limiting Rows during selection (using WHERE clause), Working with Character Strings and Dates, Using Comparison operators, BETWEEN Operator, IN Operator, LIKE Operator, IS NULL Comparison, Logical Operators, Use of Logical Operators (AND/OR/NOT Operators), Logical Operator Precedence, ORDER BY Clause, Sorting in Ascending/Descending Order, Sorting By Column Alias Name, Sorting On Multiple Columns;

Functions: SQL Functions, Types of SQL Function (Single Row/Multiple Row), Single Row SQL Functions, Character Functions (Case Conversion/Character Manipulation), Case Conversion Functions (lower (), InitCap (), UPPER ()) Character Manipulation Function (CONCAT(), INSTR(), LENGTH(), TRIM(), SUBSTR(), LPAD()), Number Functions (ROUND(), TRUNC(), MOD()), Working with Dates (LAST_DAY(), MONTHS_BETWEEN(), NEXT_DAY(), ADD_MONTHS(), ROUND(), TRUNC()) Arithmetic Operation on Dates, Date Functions and their Usage, Data type Conversion Functions, Implicit and Explicit Conversion, TO_CHAR Function with Dates, TO_CHAR Function For Numbers, TO_NUMBER and TO_DATE Functions, NVL Function and its Usage, DECODE Function and its Usage;

Grouping Records: Concept of Grouping Records and Nested Grouping, Nested Grouping of records, Group Functions, Types of group functions (MAX(), MIN(), AVG(), SUM(), COUNT()), Using AVG and SUM Functions, Using MIN and MAX Functions, Using the COUNT Function, using COUNT(*), DISTINCT clause with COUNT, Group Functions and Null Values, Using NVL Function with Group Functions, Grouping Records: Group By Clause, Grouping By More than One Column, Illegal Queries with Group By Clause, Excluding Group Results: Having Clause, Nesting Group Functions,

Sub Queries: Concept of Sub-Query, Sub Query to Solve a Problem, Guidelines for Using Sub Queries, Types of Sub-Queries (Single Row and Multiple Row) and (Single Column and Multiple Column); Single Row Sub-Query and its Execution;

Displaying Data From Multiple Tables: Concept of Join, Result of Join, Cartesian Product and Generating Cartesian Product example using Mathematical Set), Types Of Joins (EQUI, SELF, NON-EQUI, OUTER (LEFT and RIGHT)), Equi-join: Retrieving Records with Equi-join, Additional Search Conditions using AND operator, Short Naming Convention for Tables (Table Aliases), Non-Equi join and its Implementation, Outer-Join and Its Usage, Self-Join (Joining a table to Itself);

Manipulating Data of A Table/Relation: Concept of DML (Data Manipulation Language), INSERT Statement, Inserting New Rows, Inserting New Rows with Null Values, Inserting Date Values, Use of Substitution Variable to Insert Values, Copying Rows From Another Table, Update Statement to Change Existing Data of a Table, Updating Rows In A Table, Updating Rows Based on Another Table, Delete statement/ Removing Row/Rows from a Table, Deleting Rows Based on condition from another Table; Making Data Manipulation Permanent (COMMIT). Undo Data Manipulation Changes (ROLLBACK)

Database Objects: View, Table, Sequence, index, and Synonyms, DDL (Data Definition Language), Naming Convention, Creating Views, Creating Synonyms, Simple Views and Complex Views, Retrieving Data From a View, Querying a View, Modifying a View,

Including Constraints: Constraints, Concept of using Constraints, Constraint Guidelines, Defining Constraints, NOT NULL, UNIQUE KEY, PRIMARY KEY, FOREIGN KEY, FOREIGN KEY Constraint Keywords, CHECK, Adding a Constraint, Dropping a Constraint, Disabling Constraints, Enabling Constraints, Viewing Constraints, Viewing The Columns, Associated with Constraints;

Creation of a Table/Relation: CREATE TABLE Statement, Data types, The DEFAULT option, Creating Tables, Referencing Another User's Tables, Querying the Database Dictionary to view all tables in the Oracle Database, Creating a Table by Using a Sub-Query;

Managing Existing Tables and other Database Objects: The ALTER TABLE Statement, Adding a New Column in a Table, Modifying Existing Column, Dropping a Column, Renaming an Object, Truncating a Table, Adding Comments to a Table, Dropping Views, Dropping Synonyms, Dropping Tables; giving permission to other users to work on Created Tables and Revoking it (GRANT and REVOKE statement).

Informatics Practices (Code 065)

Class XI (Practical)

Duration: 3 Hours

Total Marks: 30

1. Hands on Experience

15

A problem should be given covering the following

- Table definition (The table must include constraints)
- A form with Label, Text, Command Button control, List Box, Drive List Box, Directory List Box, File List Box, Tool and Menu Bar (Any 4)
- DSN to access tables in the database
- For data connectivity (Activex Database Control)
- Change of Text box Control Properties to view Database fields

2. Practical File

05

The practical file should contain print outs from each of the following topics.

1. Create an application using Visual Basic with a Text Box control to accept a name from the user and print “Hello <Name>” in a message box. E.g. when user types his name as “Kamal Kant” in the text box and clicks OK button, a message “Hello Kamal Kant should be displayed and if he clicks on Cancel button a message as “Bye Kamal Kant” should appear.
2. Create an Application having two Text Boxes on the Window. Get Title, First Name and Last Name in it. On clicking Ok button a message should appear by joining Title + First Name + Last Name. e.g. if user enters Prof. in Title, Rajyash in First Name, and Swami as Last Name then the message to be printed should be “ Happy Deepawli Prof. Rajyash Swami”.
3. Create an application to let user guess any number and click a Play button given on the form. On clicking the Play button the application will generate a random number. If the generated number is same as guessed by the user then display a message “You Win” otherwise display a message “You Lose”.
4. Create an application to Display Image files kept in different folders in the system. The application should allow the user to navigate in the folders and list all Image Files (*.BMP, *.JPG) when ever a image file is selected it should get that picture displayed in an Image control.
5. Create an application having menu bar and tool bar to create a text file, navigate and open text files, edit text file and save changes made by the user.
6. Create a small application working as a general purpose calculator.(+, -, x, ÷)
7. SQL assignments (based on Demo Tables present in the ORACLE database for example Emp table, Dept table and SalGrade table):

- Display all the records (all columns) from table Emp.
- Display EmpNo and EName of all employees from table Emp.
- Display Ename, Sal and Sal added with Comm from table Emp.
- Display EName joined with Job with heading “Employee”, Sal*12 as “Total Salary” from table Emp.
- Display distinct Sal of employees from table Emp..
- Show the Structure of table Dept
- Write a query to display EName and Sal of Employees whose salary is greater than or equal to 3000 from table Emp.
- Write a Query to display employee name, salary and department number who are not getting commission from table Emp.
- Write a Query to display employee Number, name, sal and sal*12 as Annual Salary whose commission is not NULL from table Emp.
- Write a Query to display employee name and salary of those employee who don't have there salary in the range of 1500 to 2000
- Write a Query to display name, job, salary, and HireDate of employees who are hired between February 20, 1981, and May 1, 1981. Order the query in ascending order of HireDate.
- Write a Query to display the name and hire date of all employees who were hired in 1982.
- Write a Query to display the name, job title and salary of employee who do not have manager.
- Write a Query to display the name of employee whose name contains ‘A’ as third alphabet.
- Write a Query to display the name of employee whose name contains ‘T’ as the last alphabet.
- Write a Query to display the name of employee whose name contains ‘M’ as first alphabet ‘L’ as third alphabet.
- Write a Query to display the name of employee who is having ‘L’ as any alphabet of the name.
- Write a query to display the current system date.
- Write a Query to display employee number, name, salary, salary increase by 15% expressed as a whole number. Label the column as New Salary.
- Write a Query to display the employee’s name and salary review date, which is the date after six months of HireDate.
- Write a Query to display the employee’s name and salary review date, which is the date after six months of HireDate in format of ‘Sunday, 7 SEP, 1981’.
- For each employee display employee name and total number of weeks lapsed between HireDate and Today.
- For each employee display employee name and total number of days lapsed between Hire Date and Today.

- Create a query that produces display in the following format
<employee name> Earns \$<salary> Monthly and working as <Job >
- Write a query which displays the employee name with the first letter capitalized and all other letters lower case and length of there name string.
- Write a Query to to display the employee name and commission amount. If the employee does not earn commission, put “No Commission”.
- Write a query to display the grade of all employees based on the value of the column job as per following scheme:

JOB GRADE

PRESIDENT	A
MANAGER	B
ANALYST	C
SALESMAN	D
CLERK	E
NONE OFTHE ABOVE	O

- Write a query to display the EName and DeptNo and DName for all employees using tables Emp and Dept.
- Write a Query to display employee name, department name and location of all employees who have manager number between 7500 and 7900.
- Write a Query to display the employee name, department number and all the employees that worked in the same department as a given employee.
- Write a Query to display employee name and HireDate of employees who are employed after Employee ‘BLAKE’.
- Write a Query to display employee number, name and manager’s name with their manager number.
- Write a Query to Display the Sum, Average, Highest and Lowest salary of the employees.
- Write a Query to Display the Sum, Average, Highest and Lowest salary of the employees grouped by department number.
- Write a Query to Display the Sum, Average, Highest and Lowest salary of the employees grouped by department number and sub-grouped by job.
- Write a query to display the number of employee with same job.
- Write a query to display the average of Highest and lowest salary of each department.
- Write a query to display the difference of Highest and lowest salary of each department having maximum salary > 4000.
- Write a query to display the employee name and job for all employee in the same department as ‘ALLEN’· Write a query to display employee name and salary of those who either work in department 10 or have salary greater than employee 7521.

Before the following exercise please ensure that you are provided with a table Employee with following description

Table: Employee

Name of Column		Type
ID	NUMBER	(4)
First_Name	VARCHAR2	(30)
Last_Name	VARCHAR2	(30)
User_ID	VARCHAR2	(10)
Salary	NUMBER	(9,2)

- Use DESCRIBE command to ensure the table structure.
- Add the following data in the above Table as instructed

ID	First_Name	Last_Name	User_ID	Salary
1	Dim	Joseph	Jdim	5000
2	Jagannath	Mishra	jnmishra	4000
3	Siddharth	Mishra	smishra	8000
4	Shankar	Giri	sgiri	7000
5	Gautam	Buddha	bgautam	2000

- Populate table with first record mentioning the column list in the insert clause.
- Populate table with next two records without mentioning the column list in the insert clause.
- Populate table with 4th record and enter only ID and First_Name
- Populate table with 5th record and enter ID, User_ID, and Last_Name only.
- For record with ID = 4 update record with Last_Name User_ID and Salary.
- For record with ID = 5 update records with First_Name and Salary.
- Make the changes permanent.
- Modify the Last_Name, of the employee 3, to Gautam.
- Modify the Salary and increase it by 1000, for all who get salary less then 5000.
- Delete the employee record having First_Name as Siddharth.
- Make the changes permanent.
- Remove the entire contents of the table
- Undo the above step.
- Create a table Employee1 with columns ID, First_Name and Dept_ID from table Employee and also confirm the existence of table Employee1
- Create a view VU_Emp1 which should include column EmpNo, EName and Dept No from the table Emp.
- Create a view VU_Emp2 which should include column EmpNo, EName and Dept No from the table Emp and change the column headings as EmpNumber, Employee, Department.

- Select VIEW_NAME and TEXT from the data dictionary USER_VIEWS.
- Create the table Department table based on the following table instance chart.

Column Name	ID	Name
Data Type	NUMBER	VARCHAR2
Length	8	25

- Populate the table Department with data from table dept. Including only required columns.
- Create the table Employee based on the following table instance chart.

Column Name	ID	First_Name	Last_Name	Dept_ID
Data Type	NUMBER	VARCHAR2	VARCHAR2	NUMBER
Length	8	25	25	8

- Rename table Employee1 to Employee2.
- Drop table Employee2.
- Drop table Employee and Department
- Create table Customer as per following Table Instance Chart.

Column Name	Cust_ID	Cust_Name	Cust_Add1	Cust_Add2	Pincode	Cust_Phone
Key Type						
Nulls/Unique						
Fk Table						
Fk Column						
Datatype	number	varchar2	varchar2	varchar2	number	varchar2
Length	7	30	20	30	6	10

- Add one column Email of data type VARCHAR2 and size 30 to the table Customer.
- Change the data type of column pincode to VARCHAR2(10) in the table Customer.
- Add one more column CustomerIncomeGroup of datatype VARCHAR2(10).
- Insert few records with relevant information, in the table.
- Drop the column CustomerIncomeGroup from table Customer.
- Create table Department as per following Table Instance Chart.

Column Name	DeptID	DeptName	DeptLocation
Key Type	Primary		
Nulls/Unique		NOT NULL	
Fk Table			
Fk Column			
Datatype	NUMBER	VARCHAR2	VARCHAR2
Length	2	20	20

- Create table Employee as per following Table Instance Chart.

Column Name	EmpID	EmpName	EmpAdd	Phone	EmpSal	DeptID
Key Type	Primary Foreign					
Nulls/Unique		NOT NULL				
Fk Table						Department
Fk Column						Dept_ID
Datatype	number	varchar2	varchar2	varchar2	number	varchar2
Length	6	20	30	10	9,2	2

- Create table Employee1 as per the above Table Instance Chart but now use table level primary key addition method.
 - Create table Employee2 as per the above Table Instance Chart without any constraint while table creation.
 - Add a PRIMARY KEY constraint to the table Employee2 using the EmpID column.
 - Add a FOREIGN KEY reference on the Employee2 table that will ensure that employee records with nonexistent departments are to be prohibited.
 - Confirm that constraints were added by querying Constraint_Name and Constraint from USER_CONSTRAINTS relation.
 - Add a NOT NULL constraint to the table Employee2 on column EmpName.
 - Add a CHECK constraint to ensure, at the time of record insertion, that employee records with salary less than 2000 are to be prohibited.
 - Disable NOT NULL Constraint on the column EmpName from the table Employee2
 - Drop UNIQUE constraint from the column DeptName in table Department
8. Create an application to list all the contents of a database table using a data control object in visual basic.

9. Create an Application in Visual basic having Menu Bar tool bar and other controls to View, Add and Modify records present in the Database Tables.

3. Project **05**

The following case study is to be used to develop a team project.

A cable company in Delhi is working since 1998. They have about 2 Lac customers in different zones (North, South, East, and West). Company wants to computerise its working, which involves Customer Registration, Customer Billing, and Bill Collection on monthly basis.

Develop a Database Handling Software for the company. The software should have option to enter customer data and information of bill collection. The data entry form should also have option to navigate through the records.

The software should allow to store following information of customer and billing (Normalize this to store data in tabular form).

Customer Name
Customer Address
Customer City
Customer Zone
Customer Pin Code
Customer Phone
Customer Interest (Movies, Games etc)
Customer Monthly Income
Customer Monthly Installment
Customer Joining Date
Bill Cycle
Bill Collection Date
(Suitable assumptions can be made)

The user interfaces should be designed in visual basic and must be user friendly with correct tab order.

Note: Similar type of cases can also be encouraged, provided it should include almost every aspect of course undertaken.

4. Viva Voce **05**

Five questions from topics covered in the curriculum

Class XII (Theory)

Duration: 3 hours

Total Marks: 70

Unit No	Unit Name	Marks
1.	BUSINESS COMPUTING	10
2.	PROGRAMMING	30
3.	RELATIONAL DATABASE MANAGEMENT SYSTEM	30
		70

UNIT 1: BUSINESS COMPUTING

Introduction to Open Source based software:

Terminology: OSS, FLOSS, GNU, FSF, OSI, W3C.

Definitions: Open Source Software, Freeware, Shareware, Proprietary software, Localisation, UNICODE

Softwares : Linux, Mozilla web browser, Apache server, MySQL, Postgres, Pango, OpenOffice, Tomcat, PHP, Python

Websites: www.sourceforge.net, www.openrdf.org, www.opensource.org, www.linux.com, www.linuxindia.net, www.gnu.org.

General concepts, User interfaces (Front End), Underlying Database (Back End), Integration of User Interface and Database;

More application areas of Databases:

Inventory control, Financial Accounting, Pay-Accounting System, Invoicing Management System, Personal Management System / HRD System, Fees Management system, Result Analysis System, Admission Management System, Income Tax Management System;

Advanced Program Development Methodology: System Development Life Cycle, Relational Database Concept, Relational Database, Management System, Data Models (Entity Relationship Model), Entity and Entity Set, Attributes (Single, Composite and Multi-Valued), Relationship (One-to-One, One-to-Many and Many-to-Many), Entity Relationship Modeling Conventions, Communicating with an RDBMS using SQL, Relational Database Management System, SQL Statements, About programming language in SQL.

Data Dictionary, Data Warehousing, Data Mining, Meta Data;

Object Modeling: Introduction to object oriented modeling using Unified Modeling Language (Concepts only).

Client Server Computing: Concept of Client Server Computing.

UNIT 2: PROGRAMMING: Visual Basic

Review of Class XI;

Programming Fundamentals

Modules: Modules in Visual Basic- Form Modules, Standard Modules, and Class Modules;

Procedures: Procedures (General, Event, Function, Property);

Control Structures:

Revision of Decision Structure – IF, IF-THEN-ELSE, Select Case;

Revision of Looping Structure- Do While...Loop, Do...Loop While, For...Next, For Each...Next;

Functions: Concept of Functions, Defining and Use of User Defined functions, function to perform calculations, Parameterized Functions;

Library Functions (System Functions)

String Function: Space(), Str(), Right(), Left(), Mid(), InStr(), Len(), Ltrim(), Rtrim(), Ucase(), Lcase(), String();

Numeric Function: Sgn(), Val(), Int();

Time-Related Function: Now(), Time(), Minute(), Month();

Miscellaneous Function: MsgBox(), InputBox();

Types of forms: Single Document Interface (SDI) and Multiple Document Interface (MDI);

MDI Applications: Creating MDI form and Child form, Arranging Child Forms;

Accessing database from ORACLE using ODBC or ADO or OLEDB to connect with database.

Data Control: Accessing Data with the Data Control, Using Data-Aware Controls, Using Data Control Properties – Database Name, Exclusive, Options, Read Only, Record Source, Data Control Methods – Refresh, UpdateControls, UpdateRecord; Bound Controls: Adding Bound Text and Bound Label Controls. Data-Bound list Boxes, Grids, and Sub-Forms

ADO (ActiveX Data Objects): Connection Object, Command Object, and RecordSet Object, Special ADO Properties – Connection String (using single table), Command Text, Command Types, Cursor Locations, Cursor Types, Lock Types, Mode Types.

ADO Data Control: Simple Data linking using ADO Data Control Methods, ADO Data Control Events.

UNIT 3: RELATIONAL DATABASE MANAGEMENT SYSTEM

Review of RDBMS from Class XI

Database Fundamentals

Concept of Database Transaction, Committing a Transaction, Concept of “All or None” in a Transaction, Network Protocols Required (TCP/IP) for Data Communication, Stored Procedures, Concept of Database Fragmentation and Distributed Databases.

PL/SQL (Programming Language in SQL)

Importance of Writing Procedures, Declaring Variables: About PL/SQL, PL/SQL Block Structure, Program Constructs, Use of Variables, Handling Variables in PL/SQL, Types of Variables, Declaration, Naming Rules, Assigning Values to Variables, Initialization, and Keywords, Scalar Data types, Base Scalar Data Types, Scalar Variable Declaration, %TYPE attribute: for variable declaration, Declaring Boolean Variables, PL/SQL Record Structure, Referencing Non-PL/SQL variables, DBMS_OUTPUT.PUT_LINE;

Writing Executable Statements: PL/SQL Block Syntax and Guidelines, SQL functions in Code, SQL Functions in PL/SQL, PL/SQL Functions, Data type Conversion, Nested Blocks and Variable Scope, Operators in PL/SQL, Using Bind Variables, Programming Guidelines, Determining Variable Scope, SQL Statements in PL/SQL, Retrieving data in PL/SQL, Manipulating Data using PL/SQL, Inserting Data, Updating Data, Deleting Data, Naming Conventions, Commit and Rollback Statements, SQL Cursor, and Cursor Attributes;

Writing Control Structures: Controlling PL/SQL Flow of Execution, IF statements, IF-THENELSE Statement Execution Flow, IF-THEN-ELSIF Statement Execution Flow, Building Logical Conditions, Logic Tables, Boolean Conditions, Iterative Control: LOOP Statement, Basic Loop, FOR Loop, While Loop;

Creating Procedures: Overview of Procedures, Syntax for Creating Procedures, Developing Stored Procedures and its Advantages, Creating a Stored Procedure, Procedure Parameter Modes, Creating Procedures with Parameters, IN and OUT parameters and Usage, DEFAULT Option for Parameters, Removing Stored Procedures;

Writing Cursors: Introduction to Cursors (Implicit and Explicit), Explicit Cursor Functions, Controlling Explicit Cursors, Declaring, Opening and Closing the Cursor, Fetching data from the Cursor, Explicit Cursor Attributes (%ISOPEN, %NOTFOUND, %ROWCOUNT), controlling multiple fetches, Cursors and Records, Cursor FOR Loops, Cursor FOR Loops using Sub Queries.

Triggers: Types of Triggers: Row-Level Triggers, Statement Level Triggers, BEFORE and AFTER Triggers, INSTEAD of Triggers, Valid Trigger Type, Trigger Syntax, Combining Trigger Types, Enabling and Disabling Trigger, Replacing Trigger, Dropping a Trigger.

Development of Data Base Applications (Application Domain): Student database for school, Employee database for a company, Library Database for Library Student database management system for school, Employee database management system for a company, Library Database management system for Library, Railway Reservation System, Hotel Reservation, Inventory Control System;

Informatics Practices (Code 065)

Class XII (Practical)

Duration: 3 Hours

Total Marks 30

1. Hands on experience

15

A problem should be given covering the following features

1. Start a Standard Exe Project and it should contain MDI form with Menu Bar and Tool Bar (with Images)
2. Table structure in the database for the application with Constraints (Primary Key, Foreign Key, Check, and Unique).
3. A New Form to place an ADO component on it, for accessing data in table Stored Procedure to perform transactions/ conditional update
4. Trigger (any)
5. Making executable files of the project.

2. Records

05

1. Create an Application using Visual Basic for Students Information System Having a Student Table in Relational Database and a Student Data Form in Visual Basic to enter data into the database.
2. Create an Application using Visual Basic for Criminals Information System Having a Criminal Table in Relational Database and a Criminals Data Entry Form in Visual Basic to enter data into the database. The Data entry form should contain form level and Field level checks using procedures.
3. Create an Application using Visual Basic for Nursing Home Automation System having Linked tables (for example: Patient, Employee, Bill) in Relational Database and a required Data Entry Forms in Visual Basic to enter data into the database. The Data entry form should contain form level and Field level checks using procedures. Use of Bound Controls and Sub-Forms are to be encouraged in this application.
4. Create a database handling application for Student Expert System. Following features are to be incorporated in the application:
 - a. Create following linked tables of Student in the Relational Database.
 - i. StudentMaster : containing general information about the student.
 - ii. StudentDetail: Table to store data having details such as Class, Section, Marks and other relevant information.
 - iii. StudentFeeDetail: Should contain details like Financial Year, Class, Fee, FeeStatus(such as Paid and UnPaid)
 - iv. Accounts: General Accounts table to store fee collection details such as received from, date, chequeno and other relevant information.

- b. The database should have Procedures to update data, Insert data and to perform other database transactions.
- c. Database triggers should also be defined wherever automatic data modification is required.
- d. Visual basic forms for data entry.
- e. Procedures in Visual Basic to perform Database Transactions and Commit changes made
- f. Reporting tool to make the MIS reports, required to analyse data entry.

3. Project

05

The following case study is to be adopted for the development of project

A book publishing company B R Publishing Group is in existence since 1950. They were untouched with latest technological inventions. They are still using a traditional approach of bookkeeping and accounts maintenance.

A company, Nova technology, introduced themselves as system integrator and developers who can change existing working system into the latest concept of paper less office. They wanted few details from the company about its working. The details are as under:

- Name of the company is B R Publishing Group.
- The company is having 20 employees. One Managing Director, Two Managers (Work manager and Marketing Manager) and 17 employees who work as a team for book publishing.
- The company publishes books in different Indian languages and different topics.
- Every book involves an Author and its detail.
- The book is sold in the market at a variable discount options
- Book Seller: 30%
- Schools: 20%'
- Customer: 15%
- The company is maintaining information about Author and all its details such as Personal Information, Royalty etc.
- The company manages information about the book such as Book Name, Author, Quantity Sold, Quantity in Stock, etc.
- The company maintains Customer (Book Sellers) information. Books Sold, Subject, Language, and Amount Pending etc.
- Reports are required at different levels, such as
- Customer Listing
- Book Listing
- Language Wise Book Listing

- Topic Wise Book Listing
- Pending Amount Listing (Customer Wise, Book Wise)
- Author Royalty Detail
- Bill Generation etc.

As a developer you are required to design the project and develop it as per customer needs (Developer can also visit a publishing company to collect customer details and live data). Suitable assumptions can be made during implementation. A proper normalized database is to be maintained in the RDBMS and the front end is to be developed using advanced interface controls. User-friendly interface is to be generated.

Note: This is a sample case study. Similar type of cases can be developed on different application areas such as Library, Hospital, Transport Authority, Transporters, Wholesale Merchants, and Chemist Shops etc. The cases to be developed should preferably be obtained from live situations.

4. Viva Voce

05

Five questions from topics covered in the syllabus

Reference Books:

- Mastering Visual Basic 6 – Petroustos (BPB)
- Programming in Visual Basic 6 - Bay Ross (BPB)
- Visual Basic 6 Complete – Sybex (BPB)
- Successful Projects in Visual Basic – Christopher (BPB)
- Oracle8: The Complete Reference – George Koch, Kevin Loney (TMHP)
- Visual Basic Black Book (IDG)
- Programming in Visual Basic - McBride (BPB)
- Learn Oracle 8i – Ramalho (BPB)
- TEACH YOURSELF SQL / PL SQL USING ORACL 8i & 9i with SQL J – BAYROSS (BPB)
- Visual Basic and ORACLE SSI Press
- Oracle Programming with Visual Basic – Snowdon (BPB)
- Quan Book 'O' Level all Vol. – DOEACC (BPB)

16. MULTIMEDIA AND WEB TECHNOLOGY (Code 067)

Learning Objectives:

1. To get proficient in WEB Development using HTML/XML
2. To be able to write server & client scripts.
3. To get proficient in Web Management
4. To get proficient in creating Web Site
5. To design Graphical images using Image-Editing tools
6. To get proficient in audio & video capture and editing using software tools
7. To create and publish a self-contained multimedia CD-ROM using multimedia authoring tool.
8. To develop ability to use the Open Source Technology.
9. To develop ability to localize software applications.

Competencies:

The student will become proficient in the following:

1. Managing Self Developed web-site
2. Management of a full-fledged web portal
3. Creation & Editing of graphical images.
4. Capturing, Creating and Editing Audio and Video through external devices.
5. Embedding images & video into a presentation.

Class XI (Theory)

Duration: 3 hours

Total Marks: 70

Unit No.	Unit Name	Marks
1.	COMPUTER SYSTEM	15
2.	WEB DEVELOPMENT	25
3.	WEB SCRIPTING	20
4.	MULTIMEDIA AND AUTHORING TOOLS	10

Unit 1: Computer System

Introduction to Computer, Input Devices – Keyboards, Mouse, Joy stick, Mic, Camera; Output Devices – Monitor, Printer, Speaker, Plotter; Memory Units – Byte, Kilobyte, Megabyte, Giga byte, tera byte; Primary Memory – RAM and ROM; Secondary Storage devices – Floppy Disk, Hard disk, CD ROM, DVD, Zip Drive, DAT Drive; Power devices – UPS; Software – System Software, Application Software, Utility Software; Working on computers – switching on computer, booting computer; icons, shortcuts, taskbar, mouse pointer; typing, saving and printing a simple text file, drawing simple picture using MSPaint, using calculator option, customizing desktop, windows explorer, managing folders (creating, moving, deleting, renaming); using floppy disk drive, using CD/DVD drives; managing files (copying, moving, deleting, renaming); playing audio and video;

GUI Operating System

Important: Students/Teachers can also perform similar operation on any operating system. It is advised that the teacher while using any one operating system, give a demonstration of equivalent features for the other operating system.

Windows

General features, Elements of Desktop - Taskbar, Icon, Start button, Shortcuts, Folder, Recycle Bin, My Computer;

Start Menu: Program, Documents, Settings, Find/Search, Help, Run, ShutDown/Logoff;

Customization of Taskbar, Start menu, Display properties (Wallpaper, Font Settings, Color Settings, Screen Savers);

Program Menu: Accessories - Calculator, Notepad, Paint, Word pad, Entertainments (CD Player, Sound Recorder, Media Player, Volume Controller);

Browsers: Mozilla Firefox, Internet Explorer, Netscape Navigator;

Control Panel: Add new hardware; Add new Software, Printer Installation, Date/Time, Mouse, and Regional Settings;

Documentation

Purpose of using word processing software, opening a new/existing document, closing a document, typing in a document, saving a document, print preview, printing a document, setting up of page as per the specifications, selecting a portion of document, copying selected text, cutting selected text, pasting selected text; changing font, size, style, color of text; Inserting symbol; Formatting: Alignment – Left, Right, Center; Justification;

Unit 2: Web Development

WebPages; Hyper Text Transfer Protocol (HTTP); File Transfer Protocol (FTP) Domain Names; URL; Protocol Address; Website, Web browser, Web Servers; Web Hosting.

HTML/DHTML

Introduction, Objectives, Introduction to Universal Resource Identifier (URI) - Fragment Identifiers and Relative URI's, History of HTML, SGML, Structure of HTML/DHTML Document, Switching between opened Windows and browser (Container tag, Empty tag, Attribute);

Basic Tags of HTML: HTML, HEAD, TITLE, BODY (Setting the Fore color and Background color, Background Image, Background Sound), Heading tag (H1 to H6) and attributes (ALIGN), FONT tag and Attributes (Size: 1 to 7 Levels, BASEFONT, SMALL, BIG, COLOR), P, BR, Comment in HTML (<! >), Formatting Text (B, I, U, EM, BLOCKQUOTE, PREFORMATTED, SUB, SUP, STRIKE), Ordered List- OL (LI, Type- 1, I, A, a; START, VALUE), Unordered List - UL (Bullet Type- Disc, Circle, Square, DL, DT, DD), ADDRESS Tag;

Creating Links: Link to other HTML documents or data objects, Links to other places in the same HTML documents, Links to places in other HTML documents;

Anchor Tag <A HREF> and <A NAME>, Inserting Inline Images <IMG ALIGN, SRC, WIDTH, HEIGHT, ALT, Image Link, Horizontal Rules <HR ALIGN, WIDTH, SIZE, NOSHADE>;

Web Page Authoring Using HTML

Tables: Creating Tables, Border, TH, TR, TD, CELSPACING, CELLPADDING, WIDTH, COLSPAN, CAPTION, ALIGN, CENTER;

Frames: Percentage dimensions, Relative dimensions, Frame – Src, Frameborder, height and width, Creating two or more rows Frames <FRAMESET ROWS >, Creating two or more Columns Frames <FRAMESET COLS >, <FRAME NAME SRC MARGINHEIGHT MARGINWIDTH SCROLLING AUTO NORESIZE>, <NOFRAMES>, </NOFRAMES>;

Forms: Definition, Use – Written to a file, Submitted to a database such as MSAccess or Oracle, E-mailed to someone in particular, Forms involve two-way communication;

Form Tags: FORM, <SELECT NAME, SIZE, MULTIPLE / SINGLE> <OPTION> ... </SELECT>, <TEXTAREA NAME ROWS COLS >, , </TEXTAREA>, METHOD, CHECKBOX, HIDDEN, IMAGE, RADIO, RESET, SUBMIT, INPUT <VALUE, SRC, CHECKED, SIZE, MAXLENGTH, ALIGN>;

Document Object Model

Concept and Importance of Document Object Model, Dynamic HTML documents and Document Object Model.

Cascading Style Sheets

Introduction to Cascading Style Sheet (CSS), three ways of introducing the style sheets to your document. Basic Syntax; Creating and saving cascading style sheets. <STYLE> tag.

Examples showing the linking of external style sheet files to a document; Inline and Embed, <DIV> tag; COLOR, BACKGROUND-COLOR, FONT-FAMILY, FONT-STYLE, FONT-SIZE and FONT-VARIANT; FONTWEIGHT, WORD-SPACING, LETTER-SPACING, TEXTDECORATION, VERTICAL-ALIGN, TEXT-TRANSFORM; TEXT-ALIGN, TEXT-INDENT, LINEHEIGHT,

Introduction to Margin, Padding and Border;

MARGINS (all values), MARGIN-PROPERTY, PADDIND (all values), PADDINGPROPERTY; BORDER (all values), BORDER-PROPERTY, BACKGROUNDIMAGE, BACKGROUNDREPEAT; Additional Features, Grouping Style Sheets, Assigning Classes; Introduction to Layers, <LAYER>, <ILAYER> tag;

eXtensible Markup Language (XML)

XML: Introduction;

Features of XML: XML can be used with existing protocols, Supports a wide variety of applications, Compatible with SGML, XML documents are reasonably clear to the layperson; Structure of XML: Logical Structure, Physical Structure;

XML Markup: Element Markup i.e (<foo>Hello</foo>), Attribute Markup i.e. (<!element.name property="value">) ;

Naming Rules: used for elements and attributes, and for all the descriptors, Comments Entity

Declarations :<! ENTITY name “replacement text”>;

Element Declarations: <!ELEMENT name content>;

Empty Elements: <!ELEMENT empty.element EMPTY>;

Unrestricted Elements: <!ELEMENT any.element ANY>;

Element Content Models : Element Sequences i.e. <!ELEMENT counting(first, second, third, fourth)>,

Element Choices <!ELEMENT choose(this.one | that.one)>, Combined Sequences and Choices;

Element Occurrence Indicators :-Discussion of Three Occurrence Indicators

? (Question Mark)

* (Asterisk Sign)

+ (Plus Sign)

Character Content: PCDATA (Parseable Character data) <!ELEMENT text(#PCDATA),

Document Type Declaration (DTD) and Validation;

Developing a DTD: Modify an existing SGML DTD, Developing a DTD from XML Code, either automatically or manually;

Viewing XML in Internet Explorer, Viewing XML using the XML Data Source Object.

XSL (Extensible Style Sheet Language) or CSS (Cascading Style Sheet);

Unit 3: Web Scripting

VBScript

Introduction, Adding VBScript code to HTML page, VBScript Data type-Variant subtypes, VBScript Variables: (Declaring variable, Naming restrictions, Assigning value to variables, Scalar variables and 1-D Array), VBScript Constants, VBScript Operators, and Operator precedence;

MsgBox: functions of message box (Prompt, Buttons, Title, Helpline, Context), Return values of MsgBox function, button argument setting.

Conditional statements: If..Then.. Else, Select case;

Loops: Do loops, While.. Wend, For.. Next, For..Each..Next;

VBScript variables: Sub procedures, Function procedures;

Using VBScript with HTML form controls, Data handling functions, String functions, Date and Times functions;

Unit 4: Multimedia And Authoring Tools

Graphics Devices: Monitor display configuration, Basics of Graphics Accelerator Card and its importance;

Basic concepts of Images: Digital Images and Digital Image Representation

Image Formats :TIFF, BMP, JPG/JPEG, GIF, PIC. PDF, PSD;

Theory of design, form, line, space, texture, color, typography, layout, color harmony, unity, balance, proportion, rhythm, repetition, variety, economy, still life, light and shade, Poster Design;

Still life, colored layout, Poster Design, Designing of Books, magazines brochures, children's literature, narrative text handling, scripts in Indian Languages, picture books, comics, illustrations with photographs, scientific illustrations, conceptual illustrations, handling of assignment for the market;

Image Scanning with the help of scanner: Setting up Resolution, Size, File formats of images; image preview, Bitonal, Grey Scale and color options; Significance of PDF- creation, modification;

Animation, Morphing and Applications

Graphic Tools: Image Editing Software (Photoshop / Coreldraw)

Basic Concepts: An Introduction, creating, Opening and saving files, Menus, Toolbox, Color control icons, Mode control icons, Window controls icons; creating new images, Image capture (TWAIN)

from scanner other files;

Image Handling: Cropping an image, adjusting image size, increasing the size of the work canvas, saving an image;

Layers: Adding layers, dragging and pasting selections on to layers, dragging layers between files, viewing and hiding layers, Editing layers, rotating selections, scaling an object, preserving layers transparency, moving and copying layers, duplicating layers, deleting layers, merging layers, using adjustment layers;

Channels and Masks: Channel palette, showing and hiding channels, splitting channels in to separate image, merging channels, creating a quick mask, editing masks using quick mask mode;

Painting and Editing: Brushes palette, brush shape, creating and deleting brushes, creating custom brushes, setting brush options, saving, loading and appending brushes, Options palette;

Opacity, pressure, or exposure, paint fade-out rate, making selections, using selection tools, adjusting selections, softening the edges of a selection, hiding a selection border, moving and copying selections, extending and reducing selections, pasting and deleting selections, Image tracing (CorelDraw).

Concept of Multimedia: Picture/Graphics, Audio, Video;

Sound: Recording Sound using Sound Recorder (Capture), Sound capture through sound editing software (ex: Sound forge), Sound editing, Noise correction, Effect enhancement ;

Voice Recognition Software Philips/Dragon, MIDI Player, Sound Recorder, MONO & Stereo. Sound File Format: AIFF (Audio Input File Format from Apple Mac) , MIDI, WAV, MP3, ASF (Streaming format from Microsoft).

Importing audio and saving audio from Audio CD.

Sound Quality: CD Quality, Radio Quality, Telephone Quality;

MULTIMEDIA AND WEB TECHNOLOGY (Code 067)

Class XI (Practical)

Duration: 3 Hours

Total Marks: 30

1. Hands on Experience

15

A topic based homepage has to be developed by each student using various commands covered in HTML and VBScript

Web page should be designed with following features.

- HTML Basic Tags (html/head/title/body/B/I/U/BR/HR)
- Anchor/Image insertion/Linking
- Tables/Frame/Form
- CSS
- XML Markup / Declarations / Element Content Model
- Element Occurrence Indicators
- Buttons/Combo Box/Check Box/Text Box using VBScript

2. Practical File with following case studies

10

The practical file should be made on the following domain specific area (with supported documents and printout)

- Make a web page for Crime against Poor Community,
- Link few more pages to the developed page, containing information about Crime and Steps taken by Government. (Use HTML tags to make a Static web page)
- Use inline styling to change appearance of contents of the web page.
- Use Style sheets (embedding or linking) to change the appearance of all the pages developed in the above case.
- Enhance the above web page by providing data in sheet format.
- At this step of web page development add dynamic features such as adding time and current date to the web page
- Collect user information using forms, for registration. Display the collected user details using message box, saying thank you for registration. (Use VB Script)

Case Studies:

(These case studies can also be used to experiment the concepts learned during the course. Knowledge domain: HTML, DHTML, CSS, VB Script, and Image Editing Software's)

1. Website of a student containing personal information about student such as email address, photograph, likes, dislikes, hobbies, class, school name, achievements, favorite restra, favorite tourist places, ultimate aim of life, message to mankind, role model.

2. Website of a School providing information of a school containing Moto of school, photograph of school, brief description of school, name of the principal, facilities and infrastructure, labs, sports, faculty and departments information, results and achievements of students.
3. Website of a Restaurant providing information about types of food items, brief description about each item with pictures, price list, and availability timings.
4. Website of a Travel Agency to provide the information about various tourist places, various modes of journey available, types of hotels available.

Note:

- For developing the website collect real information from various sources.
- It is advised to break up the above-mentioned case studies into smaller modules as per coverage of the course.
- Teachers can provide alternative case studies also of similar kind.

3. Viva Voce

05

Five questions from topics covered in the curriculum

MULTIMEDIA AND WEB TECHNOLOGY (Code 067)

Class XII (Theory)

Duration: 3 hours

Total Marks: 70

Unit No.	Unit Name	Marks
1.	COMPUTER SYSTEM	05
2.	WEB TECHNOLOGIES	10
3.	WEB DEVELOPMENT	40
4.	MULTIMEDIA AND AUTHORING	15

Unit 1: Computer System

Database Terminology: Data, Record/Tuple, Table, Database

Concept of Keys: Candidate Key, Primary Key, Alternate Key, and Foreign Key;

Database Tool: Using MS– Access, Creating and Saving Table, Defining Primary Key, Inserting and Deleting Column, Renaming Column, Inserting records, Deleting Records, Modifying Records, and Table Relationship.

Unit 2: Web Technologies

Communication and network concepts

Evolution of Networking: ARPANET, Internet, Interspace;

Different ways of sending data across the network with reference to switching techniques;

Data Communication terminologies: Concept of Channel, Baud, Bandwidth (Hz, KHz, MHz) and Data transfer rate (bps, kbps, Mbps, Gbps, Tbps);

Transmission media: Twisted pair cable, coaxial cable, optical fiber, infrared, radio link, microwave link and satellite link.

Network devices: Modem, RJ45 connector, Ethernet Card, Hub, Switch, Gateway;

Different Topologies- Bus, Star, Tree; Concepts of LAN, WAN, MAN;

Protocol: TCP/IP, File Transfer Protocol (FTP), PPP, Level-Remote Login (Telnet), Internet, Wireless/Mobile Communication, GSM, CDMA, WLL, 3G, SMS, Voice mail, Application Electronic Mail, Chat, Video Conferencing;

Network Security Concepts: Cyber Law, Firewall, Cookies, Hackers and Crackers;

Introduction to Open Source based software

Terminology: OSS, FLOSS, GNU, FSF, OSI, W3C

Definitions: Open Source Software, Freeware, Shareware, Proprietary software, Localisation, UNICODE

Softwares : Linux, Mozilla web browser, Apache server, MySQL, Postgres, Pango, OpenOffice, Tomcat, PHP, Python

Websites: www.sourceforge.net, www.openrdf.org, www.opensource.org, www.linux.com, www.linuxindia.net, www.gnu.org.

Multimedia Application: Education (use of CAI tool), Entertainment , Edutainment, Virtual Reality, Digital Libraries, Information Kiosks, Video on Demand, Web Pages Video phone, Video conferencing and Health care.

Unit 3: Web Development

Review Of HTML/DHTML, VBScript covered in Class XI.

Installation and Managing WEB-Server: Internet Information Server (IIS) / Personal Web Server (PWS).

Active Server Pages (ASP): Concept of ASP, features of ASP, other equivalent tools – JSP, PHP;

Constants: String and Numeric;

Data types: Integer, Floating Point (Single, Double), String, Date, Boolean, Currency, Variant, Object;

Variables: Explicit and Implicit Declaration;

Operators:

Arithmetic: +, - (Unary and Binary), *, /, \ (integer division) mod, ^;

Comparison: <, >, <=, >=, <>, =;

Logical: AND, OR, NOT, XOR, EQV, IMP;

String Operator: & or + (for Concatenation);

Functions:

Conversion functions: Abs(), CBool(), CByte(), CInt(), CStr(), CSng(), CLng(), CDate();

String Manipulation Functions: UCase(), LCase(), Len(), Left(), Right(), Mid(), LTrim(), InStr(), RTrim(), LTrim();

Time & Date Functions: Date(), Day(), Hour(), Left(), Len(), Minute(), Month(), Monthname(), Now();

Arrays: Declaration and use of 1 dimensional arrays;

Controls: IF..THEN, IF..THEN..ELSE..END IF, IF..THEN.. ELSEIF..THEN.. END IF, SELECT..CASE..END SELECT, FOR..NEXT, FOR EACH.. NEXT, DO WHILE..LOOP, DO.. LOOP WHILE, DO UNTIL . LOOP;

Procedures and Functions, Passing parameters/arguments;

Concept of object model structure (client to server and server to client);

Objects: Properties, Methods, Events, Setting Object properties, Retrieving Object properties, calling objects/methods;

Types of Objects: Response, Request, Application, Session, Server, ASPError;

Response Object: Write Method, AddHeader, AppendToLog, BinaryWrite, Using Shortcuts <%=value/expr%>, Controlling information: Buffer, Flush Clear, End;

Request Object: Request Object Collection: QueryString, Form, ServerVariables, Cookies, ClientCertificate;

Application : Contents, Lock, Unlock, Remove, RemoveAll;

ASP Components: AD Rotator, Content Rotator, Counter, Page Counter, Permission Checker;

Text Files: Open and Read content from a text file;

Elementary Database Concepts: Concept of Table/Relation, Relationship, Candidate Key, Primary Key, Alternate Key, Foreign Key, Connecting with Databases: Creation of DSN, using OLE DB.

Working on Database: Inserting, Retrieving, Modifying/Updation of records from Tables in Databases using server objects (ADODB. Connection, ADODB. Recordset);

Server Variables: HTTP_User_Agent, REMOTE_ADDER, REMOTE_HOST, SERVER_NAME;

Unit 4: Multimedia and Authoring Tools

Movie File Formats: AVI, MPEG, SWF, MOV, DAT;

Movie Frames: Concept of Frame, Frame Buffer, and Frame Rate;

Authoring Tools; Making Animation, Embedding Audio/Video, and Embedding on the web page;

Multimedia Authoring Using Macromedia Flash

Making of Simple Flash Movie, Setting Properties, Frame Rate, Dimensions, and Background Color;

Scene: Concept of Scene, Duplicate Scene, Add Scene, Delete Scene, and Navigating between Scenes;

Layers: Concept of Layer, Layer Properties, Layer Name, Show/Hide/Lock layers, Type of Layer - Normal/Guide/Mask, Outline Color, Viewing Layer as outline, Layer Height, Adding/deleting a layer;

Frame: Concept of Frame;

Creating a Key Frame, Inserting Text Into the Frame, Inserting Graphical Elements into the frame, Converting Text/Graphics to Symbol, Inserting Symbol into the Frame, Setting Symbol Property (Graphics/Button/Movie), Inserting Blank Frame, Inserting Blank Key Frame, Inserting Key Frame into the Blank frame, Selecting all/Specific frames of a Layer, Copying/Pasting selected Frames,

Special Effects: Motion Tweening, Shape Tweening, Color effect, Inserting Sound Layer; Testing a Scene and Movie;

Import/Export (Movie/Sound and other multimedia objects)

Publishing: Publishing A Flash Movie; Changing publish Settings; Producing SWF(Flash Movie), HTML page, GIF image, JPEG Image (*.jpg), PNG Image, Windows Projector (*.exe), Macintosh Projector (*.hqx), Quick Time (*.mov), Real Player (*.smil); Testing with Publish Preview

MULTIMEDIA AND WEB TECHNOLOGY (Code 067)

Class XII (Practical)

Duration: 3 Hours

Total Marks: 30

1. Hands on Experience

15

A website based on a particular topic has to be developed by each student using various commands covered in HTML, VBScript and ASP with at least 4 web pages.

Web page should be designed with following features.

- HTML Basic Tags (html/head/title/body/B/I/U/BR/HR)
 - Functions
 - Conditional and Control Statements
 - Objects: Response/Request / Application
 - /Session /Server /ASP error
 - Image Editing using Photo Shop /Corel draw
 - Merging layers /Moving and Copying Layers
 - Use of Multimedia Authoring (Using Macromedia Flash)
- (Note: Output as Web page/Flash Movie/Windows Projector/Quick Time)

2. Practical File

05

The practical file should be made on the following domain specific area (with supported documents and printout)

- Make a Simple web page containing almost all the tags of HTML.
- Develop a Home page for Income Tax department (Simple and Textual) and store it in the directory used for Web Services on the Web-Server.
- View that web page on the Browser.
- Enhance the home page by providing links to other sample pages (e.g. Income Tax Zone, Income Tax Detail Form for an individual, Income Tax Notification, Income Tax News etc.)
- Embed Time and Date on the home page.
- Further enhance the website by providing User Registration Page. Collect the user details and Display a new web page showing Thanks For Registration. Also write appropriate functions to validate form inputs.
- Give a login facility to the user with Anonymous name and maintain the session till the User logs out.
- For user log in attempts, maintain a visitor count.
- Change the login module of the web page and now connect it to the IncomeTax User database on the server. This is to be done to store the registration detail and facilitate login to the user.
- The login page is to be made in a way that it should also provide facility to change password, if user forget password.
- Store some of the created or edited sound files on the Web-Server and provide links to play it.

- Change the appearance of the web page using pictures at appropriate places (e.g. Logo of Income Tax Department, Photograph of Income Tax Building etc.)
- Visit websites (State Govt./ Local language newspaper) and get 5 different printouts in local language.

(Note: Student can also improve the case studies from class XI and enhance it further with database and multimedia support)

3. Projects

05

Case Studies are to be divided into following parts:

Case study Part 1(Collection, Editing and Creation of Website Resources):

Create an electronic movie with various pictures, audio clipping, movie clippings, and factual text related to school / organisation;

- Introduction to 3D Animation (Using 3D Studio)
- Embedding video and audio in web pages.
- An introduction to interactive walk-through.
- Embedding walk-through into web pages.

Case Study Part 2(Development of Web Content with resources):

Case studies covered in class XI with database support with Login, Online Registration, Booking and/or ordering facility.

Sample Case Study

(Note: Other similar type of case studies can also be used for the project work)

Mr. Verma is the CEO of *copsi* soft drinks (I) Ltd. His company is having a wide network of distributors for *copsi* branded soft drinks. With the increase in sales and distribution network, it is required to adopt a new technological intervention in the existing system. He wants that the company should have a global presence over the widely popular medium, called World Wide Web. Assume that you are appointed as the senior person of the development team. You are required to collect the company information and its current requirement. For your easiness we had collected the details of the company and are as follows:

The company Information:

Name of the Company: *copsi* soft drinks (I) Ltd.

Zone: East, West, North and South

Distributors: All over the world.

Mr. Verma said that the web site should be able to reflect company in terms of :

- Home Page
- Product & Promotion Page
- Distributor Login Page / Password Recovery Page

- Distributor Specific Details Page
- Registration Page for Distributor-ship
- Company News and Flashes
- Company Profile

Technical Details:

Web site Introduction is to be made in flash.

A proper database is to be maintained for the distributor information.

Note:

- For developing the above sites/movies collect the actual information from various sources.
- It is advised to break up the above-mentioned case studies into smaller modules as per coverage of the course.
- Teachers can provide alternative case studies also of similar kind.

4. Viva Voce

05

Five questions from topics covered in the curriculum

Reference Books

- HTML Complete – Sybex (BPB)
- Mastering HTML 4 Premium Edition – Ray (BPB)
- HTML Example Book – Farrar (BPB)
- Mastering WEB DESIGNING – Maccoy (BPB)
- Inside Adobe Photoshop 6 – Bouton (BPB)
- Multimedia on the PC – Sinclair (BPB)
- Multimedia Magic – Gokul, S (BPB)
- Mastering CorelDraw 9 – Altman (BPB)
- Learn Advanced HTML with DHTML – Ramalho (BPB)
- Effective Web Design – Navarro (BPB)
- ASP, ADO and XML Complete – Sybex (BPB)
- Mastering Active Server Pages 3 – Russell (BPB)
- Practical ASP – Bayross (BPB)
- Inside Flash 5 – Kea thing (BPB)
- VBSCRIPT Interactive Course: Waite Group – Simon (BPB)
- Computer Network – A.S. Tanenbaum, (4th Edition) (PHI)
- Network Concept and Architechtures – Hancock (BPB)

Educational Qualification for the post of PGT to teach Computer Science (083), Informatics Practices (065) and Multi Media and Web Technology (067)

1. **B.E. or B. Tech** (Computer Science/IT) from a recognized University or an equivalent Degree or Diploma from an institution/university recognized by the Govt. on India.
2. **B.E. or B. Tech** (any Stream) and Post Graduate Diploma in Computers from any recognized university.
3. **M.Sc. (Computer Science) MCA or Equivalent** from a recognized University.
4. **B.Sc. (Computer Science) BCA or Equivalent** and **Post Graduate** degree in any subject from a recognized University.
5. **Post Graduate Diploma** in Computer and Post Graduate degree in any subject from a recognized University.
6. **‘A’ Level** from **DOEACC** and **Post Graduate** degree in any subject.
7. **‘B’ or ‘C’ Level** from **‘DOEACC’, Ministry of Information and Communication Technology.**